Trust, but Verify: Measuring the Reliability Envelope of a SOTA Indic ASR Model

- Trust, but Verify: Measuring the Reliability Envelope of a SOTA Indic ASR Model
  - Abstract
  - 1. Introduction
  - 2. System under test
  - 3. Challenge suite
  - 4. Metrics
  - 5. Experiments
  - 6. Results synthesis
  - 7. Claim ledger (evidence + confidence)
  - 8. Limitations
  - 9. Reproducibility
  - 10. Appendix

Trust, but Verify: Measuring the Reliability Envelope of a SOTA Indic ASR Model

> **v1.0 (2026-09-20).** Every numeric claim below links to the artifact that produced it, either committed under `code/results/` or on the research SSD (`data/challenge/` , path noted per artifact). This pass: found, fixed, and re-measured a pipeline bug that had produced a false near-total-failure claim (§5.5/F-003, real result: +9pt improvement from oracle turns, naive auto-diarization worse than nothing), reconciled an internal denominator contradiction (§5.4), re-vendored both models and found a real Core/Flex divergence on Bhojpuri LID (§5.8/F-005), ran the previously-unrun real-noise pilot and found our own synthetic transform understates real degradation ~10x (§5.6/F-006), diagnosed the Core/Flex Bhojpuri gap as a confidence/calibration difference rather than a code restriction (§5.8/F-007), added bootstrap 95% CIs to every headline number (§7), closed the OI-WER TODO (§5.7/D-006), and locked dependencies + committed the full replication bundle with checksums (§9). Remaining open items: suite expansion past 80 samples, a neural diarizer rerun of EXP-E, human semantic verification of the 19 flagged EXP-D rows, and the cause of Flex's softer Bhojpuri calibration — all noted in §8/§9 as future work, not blockers to the claims made here.

Abstract

State-of-the-art Indic ASR is no longer the dominant bottleneck: the released model scores well on its own benchmark. The bottleneck is knowing when a transcription can be trusted and why it fails. We characterize the reliability envelope of `indic-transcribe-core` and sibling `indic-transcribe-flex` on an 80-sample suite plus a 56-clip real-noise pilot. Our own synthetic noise transform, used throughout, understates real-world degradation by roughly 10x: real noise costs +20.6 WER points (95% CI [4.9,36.4], p=0.009) versus a statistically null +2.0 points from synthetic noise. Vendor-reported Hindi-belt LID absorption reproduces on Core (Bhojpuri 0/10) but not Flex (4/10, absorbed by Maithili not

Hindi) despite tied WER — absorption is model-specific. Flex mixed-mode ITN collapses currency recall 100%→0% while raising WER. A naive automatic diarizer makes multi-speaker transcription worse than no diarization (82.6% vs 66.6%). Every headline number carries a bootstrap 95% CI. Suite, runners, results released. *Scope: small, partly synthetic/real-pilot, directional.*

---

1. Introduction

- SOTA released Aug/Sep 2026; vendor benchmark OI-WER 8.7 avg on Voice of India.
- The remaining bottleneck is the layer *above* the model (see `10-thesis/THESIS.md`).
- Contributions (each must map to a §7 claim):
  1. A structured, provenance-annotated challenge suite (`50-dataset/challenge-suite-spec.md`, `50-dataset/annotation-schema.md`).
  2. Independent measurement of documented LID absorption (`F-001`, `F-002`).
  3. A negative result for interpretable reliability features (`EXP-B`) and a positive headroom result for top-K language generation (`EXP-C`).
  4. A quantified safety failure of built-in ITN for critical values (`F-004`).
  5. A retracted-and-fixed pipeline bug in the multi-speaker remedy check (`F-003`) — kept in the report as a worked example of catching a bad number before publication.
  6. A code-audit finding, not present in either vendor card, that Core and Flex diverge on default auto-LID candidate restriction (`F-005`).
  7. Evidence that this project's own synthetic noise transform, used throughout, understates real-world degradation by roughly 10x (`F-006`).
- **Not claimed:** model quality verdict, full real-world noise robustness characterization (n=56 pilot, one real corpus), code-mixing.

2. System under test

	Core	Flex
Params / base	1.2B / canary-1b-v2	1.2B / canary-1b-v2
Languages	25	27
Snapshot	`e083ab7d...` (re-vendored 2026-09-20)	`c0db6b2b...` (vendored 2026-09-20, first time)
Role	baseline (D-002)	modes/ITN (EXP-D)

Verified API surface and card-vs-code deltas: `20-models/indic-transcribe-core.md`, `20-models/indic-transcribe-flex.md`, `F-002`, `F-005`. Key code facts: 45 s hard cap, no timestamps, token log-probs via generate passthrough, Core exposes `native/mixed/romanised` though its card says native-only. **Re-vendored 2026-09-20** (was: TODO after the 2026-09-15 HF update). Both models now ship `long_form.py` / `inference.py` and a new NeMo checkpoint distribution (untested by us, needs GPU). EXP-A–D measurements were taken against the pre-update Core snapshot (`4d29eeb7a099`); EXP-E (§5.5) and the Flex second-baseline (§5.8) were re-run/run against the current

re-vendored snapshots. The diff changed `max_new_tokens`'s default (256→512, no effect: all runners pass it explicitly) and nothing else that touches our runners; `transcribe()`'s `return_lid` contract (root cause of the F-003 bug) is unchanged in the current code. New finding from the diff: **F-005** — Flex's `transcribe()` restricts default auto-LID to `RECOMMENDED_LANGS`; Core's does not. Neither card documents this. Empirically tested (§5.8): Flex recovers 4/10 on Bhojpuri where Core gets 0/10 — real, model-specific — but *not* explained by this restriction (`bho` isn't in the excluded set); the actual cause is open.

3. Challenge suite

- 80 rows = 10 per language × `{hi, ta, te, ml, kn, ur, bho, mai}`.
- 32 real clean + 24 paired synthetic background-noise + 24 paired telephone (48 synthetic, twins preserved via `clean_twin`).
- Sources: FLEURS (hi/ta/te/ml/kn/ur), Rural-Women-Bhojpuri (bho), IndicVoices (mai). Per-source licenses recorded in manifest.
- Reproduce: `docs/REPRODUCE.md`; manifest `data/challenge/manifests/exp-a-v1-80.jsonl`.

4. Metrics

Metric	Status	Note
Whitespace-token WER	used	primary metric throughout; not OI-WER (see below)
Exact match rate	used	
LID top-1 / top-K accuracy	used	
Selective prediction (AURC/AUROC/risk–coverage)	used	
Critical-value / entity surface recall	used	exact-surface only
DER	used	toy baseline only
OI-WER	done (project definition, §5.7)	normalization-tolerant + multi-reference; **not** the vendor's undisclosed OI-WER implementation

Definitions and pitfalls: `30-metrics/` (registry `30-metrics/METRICS.md`).

5. Experiments

Each subsection: Question · Protocol · Artifact · Result · Verdict · Scope limits.

5.1 EXP-A — Baseline census

- Result: 80/80 runs; WER **22.01%**; exact match 3.75%; LID top-1 **87.5%**; warnings 50%.
- Per-language WER: `te` 14.8, `bho` 14.1, `hi` 14.0, `ta` 22.4, `mai` 23.4, `kn` 24.3, `ur` 29.0, `ml` 34.1.
- Per-condition WER: clean 21.2 / noise 23.2 / telephone 22.0.

- Strongest signal: bho auto-LID **0/10** — Wilson 95% CI **[0%, 27.8%]** (n=10).
- Artifact: `results/exp-a-v1-summary.json`, CI: `code/results/stats/bootstrap.json`. Verdict: done, directional.

5.2 EXP-B — Reliability + abstention (negative result)

- 14 features (one constant removed), LOO logistic regression.
- OOF ROC-AUC **0.515** (bootstrap 95% CI **[0.387, 0.646]** — spans 0.5, consistent with "no better than chance"); AP 0.615; Brier 0.279; AURC 0.404.
- Policy: **all 80 ABSTAIN** (no cutoff met ≤10% risk @ ≥25% coverage / ≤25% @ ≥50%).
- Artifact: `results/exp-b-v1/exp-b-summary.json`, CI: `code/results/stats/bootstrap.json`. Verdict: killed; guardrail correct.

5.3 EXP-C — Adaptive LID / top-K decoding (mixed)

- top-3 truth coverage **79/80**; baseline WER 24.87 → adaptive 25.00 → oracle **22.38**.
- Paired bootstrap (baseline − oracle, n=80): mean diff **2.48 pts, 95% CI [1.04, 4.20] pts**, excludes zero, p≈0.0002 — the headroom is statistically real, not resampling noise.
- bho selected-language accuracy 0/10.
- Artifact: `results/exp-c/exp-c-summary.json`, CI: `code/results/stats/bootstrap.json`.
- Verdict: reranker killed, candidate generation alive; headroom ≈ 2.49 pts, CI-confirmed.

5.4 EXP-D — Flex native vs mixed ITN

- WER: Core native **22.01** / Flex native **22.09** / Flex mixed **33.54**.
- Paired bootstrap (Flex mixed − Core native, n=80): mean diff **+11.5 pts, 95% CI [7.8, 15.4] pts**, excludes zero, p<0.0001 — the WER regression is robust; not driven by a handful of outlier rows.
- Curated diagnostics — number surface recall 59.1/59.1/**77.3**; currency 100/100/**0**; entity 49.6/60.3/**20.5**; date 0/0/0.
- Anecdote: `₹19,500` → `₹1950` (tenfold).
- Artifacts: `results/exp-d/exp-d-summary.json`, `results/exp-d/exp-d-semantic-curated-summary.json`.
- Verdict: **do not ship Flex mixed**; surface-level only, 19 rows flagged for review.
- **Reconciled (2026-09-20):** `annotated_rows_scored: 50` and "11 number rows" are not in conflict — they're different denominators. `50` is the count of EXP-D's 80 propagated rows (32 clean + paired noise/telephone twins) carrying **at least one** annotated slot of any kind. Verified directly from `exp-d-semantic-curated-summary.json`'s row list: 11 rows carry a number slot, 3 carry a date slot, 3 carry a currency slot, 39 carry an entity slot (rows may carry more than one type, so these don't sum to 50). Practical effect: the headline **currency 100%→0%** collapse is measured on only **3 rows** — real, but thin; the **entity 49.6%→20.5%** figure rests on 39 rows and is the better-powered part of this claim. Confidence split accordingly in the claim ledger (§7, C2).

5.5 EXP-E — Diarization pilot

- **Bug found, fixed, and re-measured (2026-09-20).** `run_exp_e.py` called `asr.transcribe(...)[0]` without `return_lid=True`; since `transcribe()` returns a bare string in that case, `[0]` took the first *character*, not the transcript — the original 09-17 numbers (no-diar 99.91%, oracle-turn 99.80%, auto-VAD 100.00%) were void. Fixed and re-ran end-to-end against a live local ASR service (M5 Mac, MPS, Core re-vendored to `e083ab7d...`). Full writeup: `F-003`.
- 4 real two-speaker recs (2 hi, 2 ta; near/far field; ~60 s).
- **Real WER (2026-09-20): no-diar 66.6% / oracle-turn 57.7% / auto-VAD+KMeans 82.6%.** DER 62.51% (toy VAD/KMeans baseline, unchanged — never depended on transcript text).
- Per-sample: hindi_085 (53.4/47.4/68.4), hindi_065 (47.2/28.7/98.5), tamil_092 (74.6/73.0/73.0), tamil_076 (91.3/81.5/90.8) — no-diar/oracle/auto, all %.
- Artifact: `code/data/challenge/results/exp-e-v2/exp-e-pilot-summary.json` (SSD).
- Verdict: oracle turns give a real but modest improvement (~9 points) — not a rescue. The toy automatic diarizer makes things **worse than doing nothing** (82.6% vs 66.6%) — turn-boundary quality matters more than raw short-turn ASR weakness. Strengthens rather than removes the case for a real neural diarizer + turn padding before shipping any multi-speaker path.

5.6 EXP-F — Real-noise robustness

- **Falsifiable target met, decisively.** 56 real clips, 7 languages, from `ai4bharat/IndicVoices` (bho absent from that corpus, dropped per design). Real noise vs. real clean (unpaired, different speakers/clips): clean **13.7%** → noise **34.3%** WER, **+20.6 pts**, 95% CI [4.9, 36.4] pts, excludes zero, p=0.009. Matched EXP-A **synthetic** noise vs. clean: 21.2% → 23.2%, **+2.0 pts**, 95% CI [–5.4, +9.7] pts, includes zero, p=0.605 — not distinguishable from no effect.
- **This project's own synthetic noise transform, used throughout EXP-A/D/E, understates real-world degradation by roughly an order of magnitude.** Every "noise" condition result elsewhere in this report characterizes robustness to mild deterministic degradation only, not real acoustic conditions.
- Condition proxy: IndicVoices' `scenario` field (`Conversation`/`Extempore` = real_noise, `Read` = real_clean) — not a controlled SNR measurement; some of the gap may be spontaneous-speech disfluency rather than pure acoustic noise (see `F-006` limits).
- Artifacts: `code/results/summaries/exp-f-v1-{baseline,summary}.json`, CI: `code/results/stats/bootstrap.json` (`exp_f`, `exp_a_synthetic_noise_vs_clean`). Full writeup: `F-006`.

5.7 OI-WER re-scoring (project definition)

- No public reference implementation of the vendor's OI-WER exists, and the suite's `reference_variants` field currently holds punctuated restatements of the primary reference rather than a curated spelling/transliteration table (57/80 rows carry a variant; checked directly — they are not orthographic alternatives, see `50-dataset/annotation-schema.md`). Per the registry's documented fallback (`30-metrics/oi-wer.md`) we implemented **project OI-WER**: NFKC normalization, casefold, cross-script digit folding (Devanagari/Bengali/Tamil/Telugu/Kannada/Malayalam/Perso-Arabic digits → ASCII), punctuation stripping, and multi-reference scoring (min WER over reference ∪ variants). Decision recorded as `D-`

`006`. Implementation: `code/metrics/oi_wer.py`, tests: `code/tests/test_oi_wer.py` (10/10 pass).

- Re-scored EXP-A: mean project-OI-WER **19.90%** vs whitespace WER **22.01%** — normalization/formatting tolerance closes **~2.1 points** of the gap. Per-language: hi 11.0, bho 14.1, te 14.8, ta 16.3, kn 20.6, mai 23.4, ur 27.1, ml 31.9.
- **Verdict on C5:** the ~13-point remaining gap to the vendor's reported 8.7 average is **not** explained by whitespace/punctuation/digit-script formatting differences. This is a materially stronger basis for "vendor OI-WER does not transfer to this slice" than the original whitespace-vs-OI-WER comparison, though it still is not an apples-to-apples comparison against the vendor's own (undisclosed) OI-WER definition or their Voice-of-India corpus.
- Artifact: `code/results/summaries/exp-a-v1-oiwer-summary.json`.

5.8 Second baseline — Flex on the EXP-A suite (F-005 empirical test)

- Previously blocked (`asr_url` unset); unblocked 2026-09-20 by standing up local inference. Ran Flex through the exact same EXP-A auto-condition call Core used, same 80 samples.
- **Overall WER tied**: Core 22.01% / Flex 21.85%, paired bootstrap 95% CI on the diff **[-1.5, +1.2] pts**, includes zero, p=0.83 — not distinguishable on this suite.
- **Bhojpuri auto-LID: Core 0/10, Flex 4/10** (Wilson 95% CI [16.8%, 68.7%]). When Flex is wrong, it's absorbed by **Maithili** (5/6), not Hindi (1/6) — a different confusion pattern than Core's, not a milder version of the same one. This narrows C1/F-001's "Hindi-belt absorption" framing to Core specifically.
- This traces to a genuine code asymmetry found by re-vendoring (§2, F-005): Flex's `transcribe()` restricts default auto-LID candidates to `RECOMMENDED_LANGS` (excludes bgc/bhb only — not bho), Core's does not. That restriction does **not** explain the Bhojpuri gap (bho isn't excluded on either model).
- **Diagnosed (F-007, compute:none, reuses committed data):** LID entropy analysis of both models' own `topk` distributions shows bho is the *only* language either model shows real predictive uncertainty on (Core mean entropy 0.322, Flex 0.517 nats; every other language ≈0 for both). Core's errors are **confident, wrong commitments** (up to 99.6% probability on the wrong belt language); Flex genuinely contests bho against mai/hi with real probability mass and wins 40% of those contests. The Flex-vs-Core entropy gap itself is directional only (paired diff +0.195 nats, 95% CI [–0.083, 0.466], n=10, not significant) — the qualitative pattern (Core is overconfident-wrong, not uncertain) is the robust part. This also explains part of C6/EXP-B's negative result: a confidence-threshold reliability feature cannot catch an error mode whose signature is *high* confidence.
- Artifacts: `code/results/summaries/exp-a-v1-flex-{baseline,summary}.json`, CIs in `code/results/stats/bootstrap.json` (`exp_a_core_vs_flex`, `exp_a_flex_bho_lid`), entropy analysis: `code/results/stats/bho_lid_entropy.json` (`code/scripts/analyze_bho_lid_entropy.py`).

6. Results synthesis

- Failure surface is **language/LID**, not audio quality — **but only under synthetic degradation**. Real noise (§5.6/C8/F-006) costs +20.6 WER points, larger than any single-language WER spread in EXP-

A; the "audio quality doesn't matter much" conclusion from earlier sections does not survive contact with real conditions and should not be generalized past the synthetic transforms this project used everywhere else.

- The LID failure surface is **model-specific, not architecture-wide**: Core and Flex are tied on transcription WER but diverge sharply on Bhojpuri auto-LID (0/10 vs 4/10) with different confusion targets (Hindi vs Maithili) — see §5.8/C7.
- Obvious interpretable reliability signals are insufficient; a better reranker is required before abstention becomes useful.
- Built-in ITN is formatting without a semantic guard.
- Multi-speaker readiness is gated on turn-boundary *quality*, not just presence: oracle turns help modestly (66.6%→57.7% WER), a naive automatic diarizer actively hurts (82.6%) — re-measured 2026-09-20 after fixing a bug that had inflated the original numbers to near-total failure (§5.5/F-003).

7. Claim ledger (evidence + confidence)

#	Claim	Evidence	Confidence	Limits
C1	Bhojpuri absorbed by Hindi in auto-LID — **Core-specific**, does not generalize to Flex	EXP-A bho 0/10, Wilson 95% CI [0%, 27.8%] (Core); EXP-C top-3 contains truth 10/10; Flex gets 4/10 correct, absorbed mostly by Maithili not Hindi when wrong (§5.8, F-005)	high (n=10) for Core; the "Hindi-belt" framing itself is now known to be model-specific	1 dialect/source; two models tested, not a model-family universal
C2	Built-in mixed ITN breaks critical values	EXP-D currency 100→0 (n=3 rows), entity 49.6→20.5 (n=39 rows); WER regression bootstrap 95% CI [+7.8, +15.4] pts (n=80)	medium-high for entity/WER; **low for currency (n=3)**	surface metric, 50/80 rows carried any annotated slot — see §5.4 reconciliation
C3	Vendor's diarize-then-transcribe remedy helps modestly but is not sufficient alone; a naive automatic diarizer actively hurts	EXP-E rerun 2026-09-20: no-diar 66.6% → oracle-turn 57.7% (real WER, after fixing a `[0]`-indexing bug that had inflated the original numbers to ~99.8%) → automatic VAD/KMeans 82.6% (worse than no-diar)	high for this slice (measured end-to-end, real transcripts spot-checked)	n=4, 2 langs, DER from a toy energy-VAD/KMeans baseline
C4	Top-K generation has headroom; likelihood reranker fails	EXP-C oracle 22.38 vs 24.87; bootstrap 95% CI on the gap [1.04, 4.20] pts, p≈0.0002	medium-high	n=80, one reranker
C5	Vendor OI-WER does not transfer to this slice	EXP-A whitespace WER 22.0 / project OI-WER 19.9 vs vendor's reported OI-WER 8.7 (§5.7)	high (categorical; formatting tolerance closes only ~2 of ~13 points)	different corpus; still not the vendor's own OI-WER definition
C6	Interpretable features do not predict errors here	EXP-B AUC 0.515, bootstrap 95% CI [0.387, 0.646] (spans 0.5)	medium-high — CI confirms "no better than chance," not just a point estimate	n=80, provisional label

#	Claim	Evidence	Confidence	Limits
C7	Core and Flex are not symmetric on Bhojpuri auto-LID: Flex recovers 4/10 where Core gets 0/10, via a different confusion pattern (absorbed by Maithili, not Hindi); overall WER is tied between the two models	§5.8: paired bootstrap on 80 matched samples, WER diff CI [-1.5,+1.2]pts (p=0.83, tied); bho Wilson CI [16.8%,68.7%] vs Core's [0%,27.8%]. Code-level asymmetry (Flex-only `RECOMMENDED_LANGS` default restriction, F-005) found first but does **not** explain the Bhojpuri effect (bho not in the excluded set)	high — real inference run, both models, matched samples	cause of the Bhojpuri gap is unexplained; vendor's own 337k-clip number for the *documented* restriction (pa 0.620→0.775, unrelated to bho) not independently verified
C8	This project's own synthetic noise transform understates real-world noise degradation by ~10x — every EXP-A/D/E "noise" result characterizes mild deterministic degradation only	§5.6/F-006: real noise−clean +20.6pts CI[4.9,36.4] p=0.009 (excludes zero) vs synthetic noise−clean +2.0pts CI[-5.4,+9.7] p=0.605 (includes zero), n=28/28 real, n=24/32 synthetic, 7 matched languages	high — statistically significant real effect vs. statistically null synthetic effect	n=56 pilot, n=4/language/condition for per-language deltas; `scenario` is a proxy for real conditions, not controlled SNR; bho untested (no IndicVoices source)
C9	Core's Bhojpuri LID errors are confident, wrong commitments, not hedged uncertainty; Flex genuinely contests bho and wins 40% of the time — bho is the only language either model shows real predictive entropy on	§5.8/F-007: Core bho mean entropy 0.322 nats, Flex 0.517 (both ≈0 for every other language); paired diff CI [-0.083,0.466] (n=10, directional only)	high for the qualitative pattern (deterministic from committed topk data); low for the specific entropy-gap magnitude	n=10; *why* Flex is less sharp on bho specifically remains unexplained

Bootstrap methodology and full numbers: `code/scripts/bootstrap_cis.py` → `code/results/stats/bootstrap.json` (10,000 resamples, seed 20260920, paired bootstrap for within-sample comparisons, Wilson score interval for the n=10 binomial).

Explicitly not claimed: model is bad; noise claim false; code-mixing; any production-safety statement; DER quality of neural diarization.

8. Limitations

- n=80 (partly synthetic twins), n=4 conversations; per-language n=10 — small enough that most bootstrap CIs are wide (see §7), and the C2 currency sub-claim rests on n=3.
- Language/source/speaker confounded; no causal ranking.
- OI-WER is now measured (§5.7) but is a project-defined normalization fallback, not the vendor's own (undisclosed) OI-WER; semantic metrics remain exact-surface only.
- Single model snapshot predating the 2026-09-15 vendor update — code-level claims (F-002, D-002) have not been re-checked against it.
- EXP-E (§5.5) was re-measured 2026-09-20 after a bug fix; do not cite the pre-fix (2026-09-17) WER numbers anywhere — they are void.

- Labels use provisional `WER ≤ 0.20`; thresholds are policy choices, not learned.
- No repeated runs / decoding-seed variance measured — all point estimates are from a single decode per sample; bootstrap CIs quantify sampling uncertainty over the 80 items, not decoding stochasticity.

9. Reproducibility

- Full procedure: `docs/REPRODUCE.md`.
- Pinned model revisions, manifest, runners, and expected summary values.
- **Committed to this repo (2026-09-20):** `code/metrics/oi_wer.py` + `code/tests/test_oi_wer.py`, `code/scripts/bootstrap_cis.py` + `code/results/stats/bootstrap.json`, `code/scripts/rescore_oi_wer.py` + `code/results/summaries/exp-a-v1-oiwer-summary.json`, `code/results/summaries/exp-a-v1-flex-{baseline,summary}.json`, `code/results/summaries/exp-e-v2-pilot-{results,summary}.json`. Re-run bootstrap CIs with `code/.venv/bin/python code/scripts/bootstrap_cis.py --exp-c-hypotheses <SSD>/exp-c/exp-c-hypotheses.jsonl --exp-d-flex-results <SSD>/exp-d/exp-d-flex-results.jsonl --exp-b-decisions <SSD>/exp-b-v1/exp-b-decisions.jsonl --exp-a-core-baseline <SSD>/exp-a-v1-baseline.jsonl --exp-a-flex-baseline code/results/summaries/exp-a-v1-flex-baseline.jsonl --output code/results/stats/bootstrap.json`.
- **Closed 2026-09-20:** EXP-A/B/C/D/F per-sample and summary JSON are now committed under `code/results/summaries/` with SHA-256 checksums (`code/results/CHECKSUMS.sha256`, `shasum -a 256 -c` verified); dependencies locked (`code/requirements.lock`, `uv pip sync`).
- **Still TODO:** single `make repro` entry point chaining §7–§10; EXP-E's per-sample records (`exp-e-v2-pilot-*`) and the Flex second-baseline (`exp-a-v1-flex-*`) are already committed, but a few EXP-C/D raw SSD paths referenced inline above still assume local SSD access rather than the committed bundle — treat `code/results/summaries/` as authoritative where the two disagree.

10. Appendix

- A. Per-sample result tables (`results/*.jsonl`, SSD; CIs in `code/results/stats/bootstrap.json`).
- B. Card-vs-code delta tables (`20-models/*`).
- C. Engineering log (`90-journal/`).
- D. Known discrepancies and fixes: §5.4 EXP-D annotation-count reconciliation; §5.5 EXP-E transcribe-bug found/fixed/re-measured (F-003); §5.7 OI-WER fallback (D-006); §2/§5.8 code-audit + empirical finding F-005 (Flex-only auto-LID restriction, Core/Flex Bhojpuri divergence, C7); §5.6 real-noise pilot correcting the "audio quality doesn't matter much" synthetic-only conclusion (F-006, C8).

Generation status

Section	Source of truth	Status
§5.1–5.4	SSD summary JSON + `code/results/stats/bootstrap.json`	filled, CI-backed
§5.5 (EXP-E)	`F-003`, SSD `exp-e-v2/exp-e-pilot-summary.json`	**re-measured 2026-09-20** — bug found, fixed, real numbers in
§5.6 (EXP-F)	`EXP-F-real-noise.md`, `F-006`	**run 2026-09-20** — falsifiable target met
§5.7 (OI-WER)	`code/results/summaries/exp-a-v1-oiwer-summary.json`	filled 2026-09-20
§5.8 (second baseline)	`code/results/summaries/exp-a-v1-flex-*.json`, bootstrap.json	filled 2026-09-20, unblocked P-task from Ralph loop
§7 C1–C9	findings `F-001..007` + summaries + bootstrap CIs	filled, scope-limited; C3 re-measured 2026-09-20
Lockfile / repro bundle	`code/requirements.lock`, `code/results/summaries/`, `CHECKSUMS.sha256`	closed 2026-09-20